



# Mortgages, Subways and Automobiles

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## ABSTRACT

We examine how sustainable infrastructure investments affect household financial outcomes by studying the impact of subway expansions on mortgage repayment behavior in Delhi, India. Using administrative data from one of India's largest mortgage lenders, we find that households in postal codes with newly opened stations experience a 4.42% decline in mortgage delinquency rate and a 1.38% increase in prepayment rate. Evidence from vehicle registration records suggests that these improvements are driven by reduced automobile reliance, with vehicle spending falling by 6.52%. The effects are most pronounced among financially constrained households, who cut back on low-quality vehicle purchases and exhibit the greatest gains in repayment performance.

## 1. Introduction

Cities worldwide are expanding urban subway networks to reduce transportation-related greenhouse gas emissions (Axsen et al., 2020). As governments prioritize these investments, quantifying their societal impact becomes a first order concern. Recent studies have highlighted subways' direct benefits, such as reduced commuting costs (Severen, 2023), decreased traffic congestion (Yang et al., 2018; Anderson, 2014), and improved air quality (Gendron-Carrier et al., 2022; Chen and Whalley, 2012). Subways have also been found to generate spillover effects, such as fostering innovation (Koh et al., 2025) and firm-level productivity (Chen and Wu, 2024). However, fully capturing the benefits that subways provide is challenging because many impacts are hard to measure and unfold over the long term (Glaeser and Poterba, 2020). In this paper, we contribute to the discussion by analyzing how subway expansions in Delhi, India, affect household financial outcomes through their impact on mortgage repayments and automobile purchases.

The Delhi Metro, one of the world's largest and busiest subway systems, is widely recognized as a milestone in sustainable urban development. In 2011, it became the first subway system certified by the

United Nations to earn carbon credits, reducing greenhouse gas emissions by 630,000 tonnes annually. While its success in managing rapid urban growth and easing traffic congestion is well documented (Goel and Gupta, 2017), its potential to deliver direct financial benefits to households has received far less attention. This study shows that sustainable infrastructure investments such as the Metro can generate measurable financial benefits to households, reinforcing their dual role in advancing both environmental and economic well-being. Focusing on mortgage repayment, the largest financial liability for most households, we find that improved access to public transportation is associated with reduced automobile reliance, easing financial pressures and improving repayment outcomes. These effects are most pronounced among lower-income households, who are more likely to substitute public transit for car use, while higher-income households are less affected. The heterogeneity highlights an important equity dimension: sustainable transit investments can promote financial equity by lowering transportation costs for the most financially constrained.

Despite its importance, direct evidence on how subway access influences mortgage repayment and vehicle purchases remains limited, partly because these outcomes are shaped by many confounding factors.

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The phased expansion of the Delhi Metro offers a compelling empirical setting to address this challenge. Stations were announced years in advance and constructed on a predetermined schedule, generating plausibly exogenous variation in access. Prior research shows that most house price adjustments occur at the time of announcement rather than at station opening (Gupta et al., 2022), and that subway expansions have limited effects on local productivity and housing markets once construction is underway (Severen, 2023). By focusing on household outcomes after stations open, we minimize bias from both anticipation effects and endogenous neighborhood change. Delhi also offers rich heterogeneity in transportation costs and vehicle ownership, with a broad automobile market spanning two-, three-, and four-wheelers at varying price points. These features make it a uniquely suited setting for isolating the household-level financial effects of sustainable transit.

A second challenge is data availability, as most household financial and transportation choices are not easily observed at scale. We overcome this by combining two administrative datasets that allow us to track both repayment behavior and vehicle purchases at the individual level. The first is a unique dataset from one of India's largest mortgage lenders, covering April 2015 to February 2020. It includes detailed loan-level information, such as property location, value, monthly mortgage payments, and borrower characteristics including income and employment. The second is a comprehensive dataset on vehicle registrations from the Ministry of Road Transport and Highways, which provides vehicle type, purchase timing, and location. Linking these datasets enables us to observe how subway expansions reduce automobile expenditures and improve household financial stability. While the mortgage data captures changes in repayment behavior, the vehicle data reveals how reduced reliance on cars frees up liquidity, especially for financially constrained households.

Using the administrative data from the mortgage lender, we begin by investigating the impact of subway access on households' mortgage repayment behavior. Our key outcome variable is the delinquency or prepayment amount, defined as the difference between cumulative instalments and repayments. We classify a borrower as delinquent when the balance turns negative and as prepaid when it turns positive.<sup>1</sup> Formally, we estimate the effects of subway expansions on monthly mortgage delinquency and prepayment using a Difference-in-Differences research design. This method compares households in postal codes where new subway stations have recently opened (treated group) to those in other areas of Delhi (control group). Since subway station openings occur at different times, we employ staggered treatment periods in our analysis. Building on recent studies (Baker et al., 2022), we address potential biases from treatment effect heterogeneity using a stacked Difference-in-Differences model, which is widely adopted for its flexibility in handling variation in treatment timing and its computational efficiency.

Our empirical estimates show that subway expansions do reduce mortgage delinquency. For households in the same postal code as the new subway station, we find that the delinquency rate decreases by 4.42%, while delinquent amounts by 39.2%. However, this reduction is not immediate. According to our Difference-in-Differences event study, the reduction starts three months after the subway station opens and remains stable for up to 12 months. In contrast, subway expansions lead to a 1.38% increase in prepayment rates. Unlike the delayed response seen with delinquency, our event study shows an immediate rise in the prepayment rate during the month of subway expansions. This lasts for only few months. As a result, while there is an initial increase in prepayment amounts in the first few months following subway expansions, the average impact on prepayment amounts remains positive but

statistically insignificant 12 months after the station's opening.

To reinforce our findings, we then consider the proximity to subway stations. Specifically, we examine how the effects vary with distance by subdividing the control group into areas with postal codes that are near to and far from the treatment areas. This approach allows us to assess whether the impact of subway access decreases as the distance from the station increases. Indeed, our findings show that the positive impacts on the treatment group are more pronounced relative to control groups located further away from subway stations. In other words, among households residing outside the postal code area of a new subway station, those situated closer to the station exhibit significantly lower delinquency rates and higher prepayment rates as compared to those households located further away. This gradient or "distance-decay" effect indicates that the benefits of subway access on mortgage repayment behavior extend beyond the immediate vicinity of the stations but diminish with increasing distance.

We now turn to the underlying mechanisms driving our results. In order to test whether the observed effects are attributable to changes in automobile expenditures, we utilize government vehicle registration data. This rich dataset provides comprehensive information on all automobile transactions in Delhi during the same period as our mortgage data, including registration dates, vehicle types, prices, and ownership details. By analyzing this data, we aim to explore how the opening of new subway stations impacts private vehicle ownership patterns. To examine whether improved public transportation access reduces the reliance on private vehicles, we employ regression models consistent with our earlier baseline analysis. The central hypothesis is that as households benefit from better public transit options, their need for private vehicle ownership and the associated financial burdens diminishes, freeing up disposable income that can be redirected toward mortgage repayments, ultimately enhancing financial well-being.

Our empirical findings corroborate this hypothesis. By aggregating approximately 3 million registration records at the postal code level, we observe an overall decrease of 7 units in four-wheelers and a 2-unit decrease in two- and three-wheelers. Moreover, total automobile spending at the postal code level decreases by 6.52%, and average spending per vehicle falls by 4.69% after the opening of new subway stations. Consistent with mortgage outcomes, these effects become even more pronounced when treated areas are compared to distant postal codes rather than neighboring ones, with larger and statistically significant declines in four-wheeler registrations and vehicle expenditures. This reduction in vehicle expenditures likely reflects a reallocation of funds away from private vehicle ownership, potentially increasing the resources available for mortgage payments and easing financial constraints. An important question, then, is whether these reallocations are concentrated among households for whom transportation costs are most binding.

We address this by studying heterogeneity across households. If the mechanism operates through automobile expenditures, the impact of subway expansions should be more pronounced among financially constrained households, who are more sensitive to transportation costs than wealthier households. Because we lack income data for automobile buyers, we use vehicle quality as a proxy for income.<sup>2</sup> Households purchasing low-quality four-wheelers are more likely to be lower income and financially constrained. Following subway expansions, we find that registrations of low-quality four-wheelers decline by 9 units, suggesting that these households are opting out of purchases. In contrast, registrations of high-quality four-wheelers rise slightly (+2 units), consistent with higher-income households who do not face liquidity constraints. Overall, our results point to subway expansions reducing private vehicle expenditures and related costs such as fuel and maintenance, thereby improving the financial position of lower-income

<sup>1</sup> This definition is standard in the literature; see Jiang et al. (2014) and Gallagher et al. (2019). Mortgage prepayment refers to borrowers paying ahead on their payments. We exclude prepayments due to home sale and default, as discussed in Gerardi et al. (2023).

<sup>2</sup> Vehicles priced below (above) the average market price are classified as low- (high-) quality.

households.

To further validate this mechanism, we next turn to the mortgage data, which include applicants' initial income levels. Here, lower-middle-income households exhibit the largest gains: relative to higher-income households, they experience greater declines in delinquency rates and amounts and larger increases in prepayment rates and amounts. By contrast, the lowest-income households show no significant changes, likely because many could not afford private vehicles even before the expansions and thus realize little savings from lower transportation costs. This interpretation is consistent with additional institutional evidence on car loans in Delhi and with back-of-the-envelope calculations. We estimate that subway expansions increase monthly repayment capacity by roughly ₹11,438, equivalent to about 75 percent of the monthly installment of ₹15,000 on a typical ₹800,000 car loan, the average price of a four-wheeler in Delhi. This magnitude is sufficient to ease repayment burdens for lower-middle-income households, who are most likely to own vehicles, but relatively small compared to the disposable income of higher-income households, helping to explain why their repayment behavior is largely unaffected.

Overall, our findings show that improved public transportation infrastructure enhances household financial well-being by lowering vehicle expenditures. Vehicle costs are a major source of household debt, so this mechanism extends the literature linking liquidity constraints to higher mortgage defaults (e.g., [Ganong and Noel, 2020](#); [Ganong and Noel, 2023](#)) and lower prepayment rates (e.g., [Amromin et al., 2007](#)). We contribute to this research by demonstrating that infrastructure improvements such as subway expansions can ease liquidity constraints and improve mortgage repayment behavior, particularly among financially constrained households who are able to substitute private vehicle use with public transportation.

We then discuss alternative explanations for our findings. One possibility is that subway expansions raise household incomes or strengthen local economic conditions, indirectly improving mortgage performance. We systematically test these channels and find no supporting evidence. First, we examine the potential income effect. Subway expansions could, in principle, increase labor productivity or workforce participation, leading to higher incomes. To test the productivity channel, we compare outcomes for public- and private-sector workers. Because public-sector incomes are largely fixed, larger effects for private-sector workers would be expected if productivity had improved. We also test the labor participation channel by analyzing gender differences, since women often face greater commuting time constraints ([Farre et al., 2023](#); [Liu et al., 2025](#)). If easier commuting increased labor supply, we would expect larger gains for women. Finally, we assess the job access channel by comparing city-center and suburban borrowers, as the latter should benefit more from improved connectivity to jobs. Across all three tests, however, we find no significant differences between these groups, suggesting that the income channel is unlikely to drive our results.

Next, we examine whether subway expansions operate through local economic conditions or housing markets. If this channel is relevant, we would expect to see increases in housing values or loan amounts after station openings. However, we find no such effects, and survey data from the Reserve Bank of India (2015–2020) similarly show no rise in the share of households expecting higher housing prices. This pattern is consistent with the forward-looking nature of real estate markets, where price effects typically occur around project announcements rather than at the time of completion ([Gupta et al., 2022](#)). Housing prices often adjust at metro project announcements, with limited further response when stations open. We also test the housing price channel using loan-to-value (LTV) ratios. Higher housing prices reduce the likelihood of negative equity and strategic default, so if property value gains were driving our results, we would expect high-LTV borrowers to become less delinquent. However, we observe no such pattern. These findings align with recent evidence showing that subway expansions have limited effects on local productivity and housing markets ([Severen, 2023](#)).

Finally, we conduct a series of robustness checks to address potential

threats to our identification strategy. To assess the role of anticipation effects, we perform placebo regressions using pseudo-treatment dates prior to subway openings and find no evidence of behavioral responses. This helps ensure that the observed effects are driven by the actual opening of stations rather than forward-looking behavior. Google Trends data confirm this pattern, showing that public attention remained low during construction and rose only once stations became operational. We also validate the parallel trends assumption using the sensitivity analysis of [Rambachan and Roth \(2023\)](#), which shows that our estimated effects remain stable even under sizable deviations from parallel trends. Additional checks find no evidence of confounding from macroeconomic shocks, alternative data transformations, and borrower risk heterogeneity. Across all specifications, the results indicate that subway expansions improved mortgage performance, consistent with the mechanism that reduced vehicle expenditures ease household financial constraints.

Our findings carry important policy implications as governments focus on sustainable transport infrastructure development. While sustainability initiatives can bring short-term financial pressures like higher taxes and energy costs, our research shows they can also deliver financial benefits to households. By lowering transportation costs and freeing up funds, subway expansions support better mortgage repayment, revealing an overlooked advantage of sustainable infrastructure. Neglecting these spillover effects will underestimate the full economic value of subway infrastructure. Our back-of-the-envelope estimates suggest that by reducing delinquency, new subway stations could lower mortgage default rates by 0.08%. Thus, a reduction in defaults not only strengthens household balance sheets but also contributes to financial sector stability, supporting long-term economic growth. These findings demonstrate that transit infrastructure investments could be viewed as both environmental and financial interventions, with the potential to enhance economic resilience and inclusion.

### 1.1. Related literature

This paper contributes to several strands of literature. First, it adds to the literature on the effects of subways. Previous studies have shown that subways can reduce air pollution ([Chen and Whalley, 2012](#); [Gendron-Carrier et al., 2022](#); [Li et al., 2019](#)), road congestion ([Gu et al., 2021](#); [Yang et al., 2018](#); [Anderson, 2014](#)), and commuting costs ([Severen, 2023](#)) in different cities. There could also be higher firm-level productivity ([Chen and Wu, 2024](#)). On the other hand, the growing investment in subway networks may have negative spillover effects including increases in urban inequality ([Lee and Tan, 2024](#)) and coronavirus infection ([Harris, 2020](#)). In terms of housing, much attention has been given to the role of urban transport in property value appreciation (e.g., [Zheng et al., 2016](#); [Gupta et al., 2022](#)). Here, we focus on mortgages, a relatively unexplored area. Our paper adds to the literature by focusing on the effects on households' mortgage repayment decisions. By documenting the spillover effects of subway expansions in improving households' financial flexibility through decreased auto purchases, we highlight a previously unexplored mechanism.

Next, this study is related to the literature on mortgage delinquency. While studies have highlighted the role of income falsification ([Jiang et al., 2014](#)), immigrants ([Lin et al., 2016](#)), health insurance ([Gallagher et al., 2019](#)), and house price ([Haughwout et al., 2008](#)) on mortgage delinquency, liquidity shocks have often been cited to play a key role in mortgage delinquency. [Kaufmann et al. \(2011\)](#) find that higher energy expenditures increase delinquency rates, while [Low \(2023\)](#) identify liquidity shocks as key triggers of default. Consistently, [Ganong and Noel \(2023\)](#) attribute 70% of mortgage defaults to cash-flow shocks alone. This paper contributes to the growing body of research emphasizing the role of financial constraints in mortgage delinquency. By connecting delinquency with vehicle purchases after subway expansions, we provide an additional channel through which vehicle expenditures can influence mortgage delinquency.

Finally, we contribute to the literature on mortgage prepayment. Unlike mortgage delinquency, relatively little work has been done on the factors that influence mortgage prepayment. Existing papers focus on the effects of monetary policy (Berger et al., 2021; Gerardi et al., 2023) and liquidity constraints (Amromin et al., 2007) on mortgage prepayment. While prepayment and delinquency may be influenced by similar factors, few studies directly link them. Scharlemann and Shore (2022), using interest rate resets under the HAMP program, show that higher mortgage rates increase default and delinquency, while prepayments remain limited due to weak refinancing incentives. Our paper contributes to the literature on prepayment by highlighting the role of vehicle expenditures on mortgage prepayment instead of focusing on monetary policy. Importantly, this paper establishes a connection between mortgage delinquency and prepayment by documenting the contrasting effects of subway expansions on these two aspects.

The remainder of the paper is organized as follows. Section 2 provides background on Delhi's subway expansions, outlines key features of India's mortgage market, and describes the data used in this study. In Section 3, we present our main empirical analyses on mortgage delinquency and prepayment. Section 4 examines the role of automobile purchases. Section 5 tests for other channels, conducts several robustness checks, and presents a simple back-of-the-envelope exercise to quantify the benefit of fewer delinquency. Finally, Section 6 concludes.

## 2. Background

In this section, we first provide the institutional background on subway expansions in Delhi, followed by an overview of the mortgage market in India. We then describe the mortgage and vehicle registration data utilized in this study.

### 2.1. Subway expansions in Delhi

Delhi, the capital city of India, has a population exceeding 33 million as of 2024. This places it comparable in size to countries like Peru and Malaysia, while surpassing the populations of nations such as Australia and the Netherlands. In India, the Delhi Metro is the largest and busiest subway rail system, operating over 3,000 trips daily and carrying approximately 2.5 million passengers. The Delhi Metro also features interchanges with the Rapid Metro Gurgaon and Noida Metro, enhancing connectivity across the region. It is managed by the Delhi Metro Rail Corporation, which is jointly owned by the governments of India and Delhi. It commenced operations on 25 December 2002 and has since undergone several expansions. To date, phases 1, 2, and 3 were completed in 2006, 2011, and 2021, respectively. These development phases represent major milestones, providing Delhi and the National Capital Region with a reliable, efficient transportation system.

The impact of the Delhi Metro has been well-recognized. It has helped to ease the immense pressure on Delhi's road networks, reducing traffic congestion and the travel time for commuters (Jain et al., 2014). Moreover, the expansion of the metro network has been pivotal in tackling the city's growing environmental issues, such as air pollution and greenhouse gas emissions. By offering an attractive alternative to private vehicles, the Metro has played a key role in lowering vehicular emissions, directly addressing the critical problem of deteriorating air quality (Goel and Gupta, 2017). Globally, the Delhi Metro's impact on reducing greenhouse gas emissions has been recognized through carbon credits awarded under the Kyoto Protocol's Clean Development Mechanism (CDM). This distinction underscores the Metro's critical role in advancing sustainable urban transport and addressing environmental

challenges.

In our study, we focus on the expansion of subway stations in the Delhi Metro between 2015 and 2019.<sup>3</sup> This period captures the majority of station openings during Phase 3 of the expansion, which introduced new subway stations in 41 different postal code areas across Delhi. A postal code in India consists of a six-digit numeric code used by the Indian Postal system. In total, there are 100 postal codes in Delhi, ranging from 110000 to 110099. On average, each postal code covers an area with a radius of approximately 2.18 km. Fig. 1 shows the evolution of subway stations in Delhi from 2015 to 2019. The blue circles represent subway stations that existed up to 2014, while the red circles indicate new subway lines constructed between 2015 and 2019. Grey circles denote stations built either outside of Delhi or after 2019.

Planning for Phase 3 began well before the opening of the stations. According to the Performance Audit Report by the Comptroller and Auditor General of India (CAG), the Delhi Metro Rail Corporation (DMRC) initiated preparation of the Detailed Project Report (DPR) in 2008, with the first version submitted in March 2010. A revised DPR was completed in February 2011, following a traffic study by RITES (formerly Rail India Technical and Economic Service) and feedback from the Government of the National Capital Territory of Delhi (GNCTD). Final approval for implementation was granted by the DMRC Board, GNCTD, and central authorities between March and August 2011.<sup>4</sup> These approvals marked the first authoritative public disclosure of the project's scope and alignment. Construction then proceeded over the next several years, with station openings occurring between 2015 and 2019. This timeline suggests that households were aware of the forthcoming subway well before the stations became operational, and any long-term anticipation effects such as increases in housing prices would likely have realized prior to station openings.

Phase 3 of the Delhi Metro expansion aimed to extend the metro's reach and improve connectivity within Delhi and its growing suburbs. This phase added over 160 km of new lines, including the Magenta, Pink, and Grey Lines, along with extensions to the Blue, Pink, Magenta, and Violet Lines. The project also integrated the Rapid Metro Link in Gurgaon, enhancing transit options between Delhi and its satellite cities. By offering a sustainable alternative to road travel, Phase 3 addressed Delhi's increasing transportation demands. It improved access to underserved areas, making commuting more efficient for millions of residents. The expansion was a key step in the city's efforts to develop a modern transit system that reduces pollution and eases traffic congestion. Beyond improved connectivity, the Metro has the potential to enhance residents' quality of life in various ways. Given these developments, it is of interest to examine how subway station openings between 2015 and 2019 during Phase 3 impacted household finances.

### 2.2. India's mortgage market

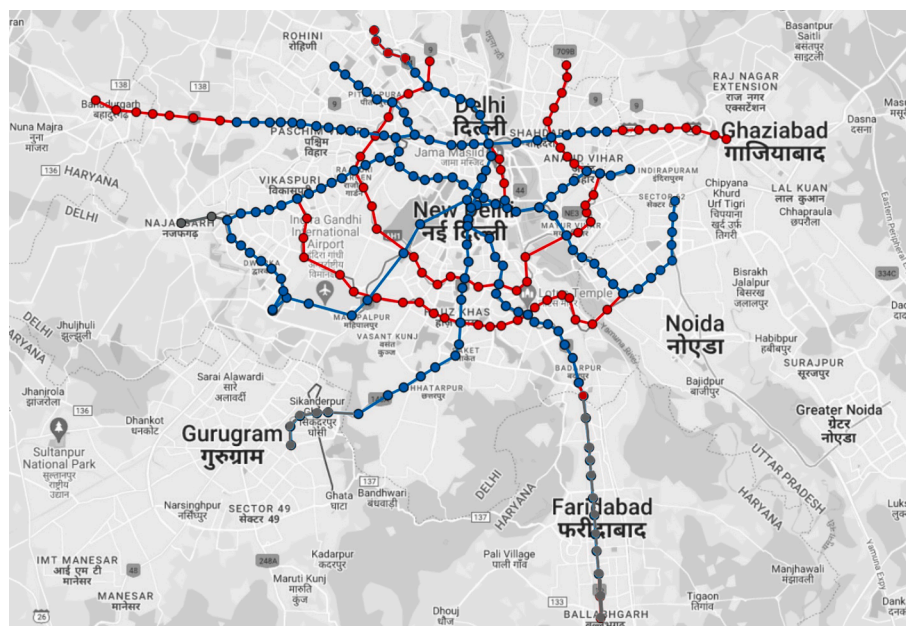
Residential mortgage loans are the primary form of household credit in India. As of December 2014, outstanding residential mortgages totaled approximately ₹6.0 trillion, accounting for 55.7% of total household debt.<sup>5</sup> These mortgage products are typically structured as either floating-rate or fixed-rate loans, each with distinct implications for borrower behavior. Fixed-rate loans in India generally offer an initial fixed rate that resets every two years. During periods of market

<sup>3</sup> Here, subway expansions refer to the opening of new subway stations, i.e., the point at which the infrastructure becomes operational and accessible to commuters rather than the start of construction. In Delhi, the construction period typically lasts 3–4 years.

<sup>4</sup> For more details, see "Report No.11 of 2021 - Performance Audit on Implementation of Phase-3, Delhi Mass Rapid Transit System by DMRC", Ministry of Housing and Urban Affairs, Dec 2021.

<sup>5</sup> For more details, see "Quarterly BSR-1: Outstanding Credit of Scheduled Commercial Banks", The Reserve Bank of India (RBI), 2014.





**Fig. 1.** Subway expansions in Delhi. *Notes:* This figure presents the evolution of the subway stations in Delhi from 2015 to 2019. The blue circles represent existing subway stations till 2014, while the red circles refer to new subway lines that were built from 2015 to 2019. The grey circles are the stations built outside Delhi or after 2019. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

volatility, lenders may revise the interest rate, and borrowers who disagree with the new terms retain the option to pre-close the loan.

In contrast, floating-rate mortgages are directly linked to the lender's base rate, with interest calculated on a daily reducing balance. These rates fluctuate with market conditions, and lenders may adjust the Equated Monthly Installment (EMI), loan tenure, or both in response to interest rate changes, even without a formal change in the base rate. These loans feature dynamic adjustment mechanisms, such as periodic changes to the EMI or loan tenure, offering borrowers more flexibility in managing repayment. Reflecting these advantages, floating-rate mortgages have long dominated the market, comprising about 75% of all outstanding home loans as of 2012.<sup>6</sup> More recent industry estimates suggest that this share has risen to 80–85% of the ₹33 trillion home loan market, underscoring borrowers' strong preference for repayment flexibility and potential savings on interest costs.<sup>7</sup>

Given the prevalence of floating-rate mortgage products, mortgage prepayment is common and institutionally supported in India. In 2012, the Reserve Bank of India (RBI) mandated the removal of prepayment penalties on floating-rate mortgages. Therefore, mortgage borrowers were able to prepay their mortgage without incurring additional costs. Additionally, media coverage and public financial advice have consistently framed prepayment as a prudent strategy to reduce long-term interest expenses.<sup>8</sup> Consequently, mortgage borrowers in India face few institutional or informational constraints when opting to accelerate loan repayment.

### 2.3. Mortgage data

To examine the impact of subway expansions on household financial well-being, we use a combination of two datasets. The first is mortgage data from one of India's largest mortgage lenders, which holds over 20

percent of the market share in India and is representative across all states. This rich administrative dataset contains detailed information on borrowers, loan types, and mortgage repayments. For each mortgage applicant, we have demographic data such as gender, age, occupation, and annual income. For the loans, we have information of loan tenure, interest rate type (fixed or floating), property value at purchase, loan amount, borrower address, and property postal code. Importantly for our analysis, the dataset includes panel-level information on monthly repayments, such as the installment start date and the amount of each monthly payment.

Our dataset begins in April 2015, coinciding with the start of Phase 3 of the Delhi Metro expansion.<sup>9</sup> We exclude data from March 2020 onwards due to the onset of the coronavirus pandemic, which led to lockdowns and shifting commuting patterns as more people began working from home. For the period between April 2015 and February 2020, we track monthly indicators for delinquency and prepayment, as well as the corresponding amounts for each mortgage applicant. Formally, loan delinquency occurs when a borrower fails to make timely payments in accordance with the terms of the loan agreement. In our dataset, borrowers are considered delinquent if they fall behind on mortgage payments, and the delinquency amount is defined as the cumulative installments owed minus the total amount paid. Conversely, mortgage prepayment refers to paying off a loan before the scheduled maturity date. Prepayment includes making extra payments, exceeding the required monthly installment, or fully repaying the remaining loan balance in a lump sum. Borrowers are deemed to have prepaid if their account balances exceed zero at the end of the month. Similar to delinquency, the prepayment amount is measured as the total repayment minus the cumulative installment amount.<sup>10</sup>

When constructing our main sample, we focus on the group of households who purchase their properties before the expansions of the

<sup>6</sup> For more details, see "Report of the Committee to Assess the Feasibility of Introducing More Long-Term Fixed Interest Rate Loan Products by Banks", RBI, 2012.

<sup>7</sup> See "Why prepaying your home loan in 2025 is a smart financial move amid high interest rates and rising living costs", MintGenie News, 2025.

<sup>8</sup> For instance, see "Prepayment of home loan", The Economic Times, 2011.

<sup>9</sup> The initial set of stations in our study began operations in June 2015, while the last group of stations opened in October 2019.

<sup>10</sup> Since both delinquency amount and prepayment amount are cumulative, it is possible to see a large number of them. Additionally, prepayment amount could be much larger than delinquency amount as borrowers are likely to prepay entire remaining balance.

subway. This is to reduce the possibility of selection bias that may occur if individuals purchase houses due to their preference for such transit option. Table 1 presents the summary statistics, including the number of observations, the means, and the standard deviations of the full sample, the control sample, and the treated sample from April 2015 to February 2020. In all, we have 9,681 households in our sample, with 45 percent of them being in the treatment group and 55 percent of them in the control group.

There are several key observations regarding the monthly installment payments. On average, monthly installments amount to ₹24,271.<sup>11</sup> In addition, the data reveal a high delinquency rate, with 29% of borrowers falling behind on their payments each month. When we compare the treatment and control groups, we observe a slightly higher delinquency rate in the control group (30%) compared to the treatment group (28%). However, the delinquency amounts tell a different story: the average delinquency amount in the control group is lower than that of the treatment group. This difference is likely driven by substantial variation in payment behaviors, as shown by the standard deviation in the treatment group, which is more than three times that of the control group.<sup>12</sup> Despite the high delinquency rate, the prepayment rate is also high. The overall monthly prepayment rate stands at 60%. Thus, a rough interpretation of our summary statistics suggests that 29% of borrowers are delinquent, 60% are prepaying, and only 11% are right on track.<sup>13</sup>

Table 1 also reports the main time-invariant mortgage and demographic characteristics. The proportion of mortgage with fixed interest rate is 14% within the full sample, and the average loan tenure is 7,632 days (about 21 years). The loan to value ratio is measured as the ratio of the total mortgage amount to the appraised value of the property. The average LTV ratio in our sample is 56%. Regarding demographic characteristics, 69% of individuals in our sample are male. The average age of mortgage applicants is 49 years, and 65% are employed in the private sector. Data on annual income is more limited, with only 6,761 observations, and the average annual income is ₹413,053. On average, the variables between the treatment and control group are close to one another.

#### 2.4. Vehicle registration data

The second dataset utilized in our analysis is the vehicle registration data, provided by the Ministry of Road Transport and Highways in India. This dataset comprises over 3 million records of vehicle registrations in Delhi, spanning from April 2015 to February 2020. The data includes detailed information such as the registration date, the district-level Regional Transport Office (RTO) where the vehicle is registered, the vehicle manufacturer, body type, price, and vehicle category. Table 2 provides the summary statistics, presenting the number of observations, mean values, and standard deviations for the entire sample, as well as for the control and treated samples, during the specified period.

According to Table 2 Panel A, 74% of vehicle registrations in Delhi are for two- and three-wheelers, while four-wheelers account for the remaining 26%. Since both two- wheelers and three-wheelers are primarily used for short-distance commuting in Delhi, we group them together for our analysis. The average price of two- and three-wheelers is ₹68,376, considerably lower than the average price of four-wheelers

which stands at ₹840,663.<sup>14</sup> The distribution of automobile types is relatively similar between the control and treatment groups. In the control group, 75% of registered vehicles are two- and three-wheelers, compared to 74% in the treated areas. The average price of two- and three-wheelers is ₹66,246 in the control group, compared to ₹70,255 in the treatment group. For four-wheelers, the average price is ₹813,157 in the control group, slightly lower than ₹863,521 in the treatment group.

Next, we aggregate our data at the postal code level. Table 2 Panel B presents the descriptive statistics. On average, 415 two- and three-wheelers and 143 four-wheelers are sold in each postal code area. We note that households in areas with subway stations purchase more vehicles on average relative to those in control areas. In the control group, an average of 337 two- and three-wheelers are registered each month, compared to 512 in the treatment group. Similarly, the treatment group registers more four-wheelers on average (184) than the control group (112). While different factors like population or income differences could explain some of these variations, they do not undermine the suitability of these groups in our analysis, as our focus is on changes in vehicle registration.

We further categorize the automobiles into high-quality and low-quality based on whether their prices are above or below the mean within each category. This follows existing literature, such as Bai et al. (2020), which shows that the quality of an automobile is strongly correlated with its price. Overall, there are 85 high-quality and 330 low-quality two- and three-wheelers. In contrast, the average number of four-wheelers is 41 high-quality and 102 low-quality units. We then examine the distribution between the control and treatment groups. In control areas, 70 out of 337 two- and three-wheelers sold are high-quality, compared to 106 out of 512 in the treated areas. Similarly, 31 out of 112 four-wheelers in control areas are high-quality, versus 55 out of 184 in the treated areas. We also analyze average and total spending on vehicles, presenting these figures in logarithmic form, and find that spending levels are similar between control and treatment groups at the postal code level.

### 3. Effects of subway expansions on mortgage repayments

The main objective of this paper is to study the effects of subway expansions on mortgage delinquency and prepayment in Delhi. Specifically, we aim to address two key questions: first, whether the introduction of subway stations influences households' monthly delinquency rates and amounts, as well as mortgage prepayment activity; second, what mechanisms drive these effects. Similar to Koh et al. (2025), our focus is on subway extensions rather than the initial opening of subway lines. An advantage of examining these extensions is that the new stations often function as linking hubs, which makes their location choices relatively exogenous and helps mitigate potential endogeneity bias. Furthermore, subway stations were planned many years in advance, making it unlikely that their openings coincided with improvements in local economic conditions.

As discussed in section 2.1, the phased expansions of the Delhi Metro offer a quasi-experimental setting to assess their impact on households. However, the staggered opening of new subway stations divides the sample into already-treated, not-yet-treated, and never-treated groups. A key concern with staggered Difference-in-Differences (DiD) is the potential for biased comparisons between late- and early-treated units, which can distort Two-Way Fixed Effects (TWFE) estimates when treatment effects vary across cohorts (Baker et al., 2022). Given the variation in station sizes and locations across expansion areas, treatment effects are likely to be heterogeneous, increasing the risk of bias in TWFE estimates. To address this issue, we adopt a stacked DiD model, an approach that has become increasingly common in recent empirical

<sup>11</sup> Note that ₹ refers to the symbol of the Indian rupee. Based on the exchange rate on February 2020, ₹24,271 is equivalent to around 340 US dollars.

<sup>12</sup> These differences should not be a concern, since our DiD specification relies on changes over time rather than on baseline levels.

<sup>13</sup> The delinquency rate in our sample is higher than the 19% home delinquency rate reported by Gallagher et al. (2019) in the U.S. market. Meanwhile, borrowers in our sample exhibit relatively high rates of mortgage prepayment. As discussed above, this is due to the bank in our study complying with the RBI's directive to eliminate all prepayment penalties from 2012 onward.

<sup>14</sup> Based on the exchange rate on February 2020, ₹68,376 is equivalent to around 957.3 US dollars. ₹840,663 is equivalent to about 11,769.3 US dollars.

**Table 1**  
Summary statistics.

|                                       | Full Sample |         |           | Control |         |           | Treated |         |           |
|---------------------------------------|-------------|---------|-----------|---------|---------|-----------|---------|---------|-----------|
|                                       | Obs.        | Mean    | St.Dev.   | Obs.    | Mean    | St.Dev.   | Obs.    | Mean    | St.Dev.   |
| <b>Panel A: Mortgage Variables</b>    |             |         |           |         |         |           |         |         |           |
| Delinquency                           | 361,598     | 0.29    | 0.45      | 177,054 | 0.30    | 0.46      | 184,544 | 0.28    | 0.45      |
| Delinquency Amount (₹)                | 361,598     | 7,212   | 76,414    | 177,054 | 5,581   | 31,680    | 184,544 | 8,777   | 102,340   |
| Prepayment                            | 361,598     | 0.60    | 0.49      | 177,054 | 0.58    | 0.49      | 184,544 | 0.62    | 0.48      |
| Prepayment Amount (₹)                 | 361,598     | 84,420  | 373,027   | 177,054 | 69,264  | 286,196   | 184,544 | 98,963  | 440,040   |
| Monthly Installment (₹)               | 361,598     | 24,271  | 88,764    | 177,054 | 21,865  | 64,955    | 184,544 | 26,579  | 106,675   |
| Fixed Interest Rate                   | 9,681       | 0.14    | 0.34      | 5,385   | 0.11    | 0.32      | 4,296   | 0.17    | 0.37      |
| Loan Tenure (day)                     | 9,681       | 7,632   | 2,069     | 5,385   | 7,636   | 2,168     | 4,296   | 7,629   | 1,938     |
| Loan to Value Ratio                   | 9,681       | 0.56    | 0.25      | 5,385   | 0.58    | 0.25      | 4,296   | 0.54    | 0.24      |
| <b>Panel B: Demographic Variables</b> |             |         |           |         |         |           |         |         |           |
| Male                                  | 9,681       | 0.69    | 0.46      | 5,385   | 0.68    | 0.47      | 4,296   | 0.70    | 0.46      |
| Age                                   | 9,681       | 49      | 10        | 5,385   | 48      | 10        | 4,296   | 50      | 10        |
| Private Company                       | 9,681       | 0.65    | 0.48      | 5,385   | 0.64    | 0.48      | 4,296   | 0.67    | 0.47      |
| Annual Income (₹)                     | 6,761       | 413,053 | 1,563,601 | 3,782   | 395,791 | 1,030,151 | 2,979   | 434,967 | 2,049,765 |

*Notes:* This table reports the means and standard deviations of mortgage and demographic variables using the full sample, the sample in the postal codes without subway expansions (Control), and the sample in the postal codes with subway expansions (Treated) from April 2015 to February 2020. Variable *Fixed Interest Rate* is an indicator for individuals whose mortgage interest rate is fixed and variable *Private Company* is an indicator for those who do not work for public sector or government.

**Table 2**  
Summary statistics – vehicle.

|                                               | Full Sample |         |           | Control   |         |           | Treated   |         |           |
|-----------------------------------------------|-------------|---------|-----------|-----------|---------|-----------|-----------|---------|-----------|
|                                               | Obs.        | Mean    | St.Dev.   | Obs.      | Mean    | St.Dev.   | Obs.      | Mean    | St.Dev.   |
| <b>Panel A: Transaction Level</b>             |             |         |           |           |         |           |           |         |           |
| Two- and Three-Wheeler                        | 3,191,266   | 0.74    | 0.44      | 1,483,656 | 0.75    | 0.43      | 1,707,610 | 0.74    | 0.44      |
| Four-Wheeler                                  | 3,191,266   | 0.26    | 0.44      | 1,483,656 | 0.25    | 0.43      | 1,707,610 | 0.26    | 0.44      |
| Price of Two- and Three-Wheeler (₹)           | 2,374,082   | 68,376  | 1,457,884 | 1,112,770 | 66,246  | 200,628   | 1,261,312 | 70,255  | 1,991,237 |
| Price of Four-Wheeler (₹)                     | 817,184     | 840,663 | 1,039,530 | 370,886   | 813,157 | 1,019,981 | 446,298   | 863,521 | 1,054,956 |
| <b>Panel B: Postal Code Level</b>             |             |         |           |           |         |           |           |         |           |
| Two- and Three-Wheeler (count)                | 5,722       | 415     | 490       | 3,303     | 337     | 412       | 2,419     | 512     | 563       |
| Two- and Three-Wheeler (High Quality) (count) | 5,722       | 85      | 102       | 3,303     | 70      | 87        | 2,419     | 106     | 117       |
| Two- and Three-Wheeler (Low Quality) (count)  | 5,722       | 330     | 396       | 3,303     | 267     | 333       | 2,419     | 415     | 454       |
| Four-Wheeler (count)                          | 5,722       | 143     | 127       | 3,303     | 112     | 113       | 2,419     | 184     | 133       |
| Four-Wheeler (High Quality) (count)           | 5,722       | 41      | 41        | 3,303     | 31      | 36        | 2,419     | 55      | 43        |
| Four-Wheeler (Low Quality) (count)            | 5,722       | 102     | 93        | 3,303     | 81      | 82        | 2,419     | 129     | 101       |
| Average Spending (log)                        | 5,722       | 13      | 0.66      | 3,303     | 12      | 0.73      | 2,419     | 13      | 0.53      |
| Total Spending (log)                          | 5,722       | 18      | 1.59      | 3,303     | 18      | 1.86      | 2,419     | 19      | 0.76      |

*Notes:* This table reports the means and standard deviations of variables at transaction level (Panel A) and postal code level (Panel B) using the full sample, the sample in the postal codes without subway expansions (Control), and the sample in the postal codes with subway expansions from April 2015 to February 2020. Variables are the monthly registration indicator for two- and three-wheeler, indicator for four-wheeler, price of two- and three-wheeler, and price of four-wheeler at individual level.

work (e.g., [Deshpande and Li, 2019](#), [Cengiz et al., 2019](#)).<sup>15</sup> This method constructs event-specific datasets that include all treated observations within a cohort and all never-treated units within the relevant treatment window. Treatment effects are then estimated separately for each stack and combined using variance-weighted averages across stacks.

### 3.1. Baseline specification

Formally, we estimate the impact of subway expansions on household mortgage delinquency and prepayment using a stacked Difference-in-Differences (DiD) model, based on subway station openings in 41 postal code areas in Delhi. Our baseline control group consists of the remaining 59 postal code areas where no subway stations opened between 2015 and 2019. Thus, each of the 41 stacks in the DiD model compares the monthly changes in mortgage delinquency and prepay-

ment in treated postal code areas before and after subway expansions with the corresponding changes in the control areas. The regression specification is as follows:

$$Y_{ijm} = \sum_{m=\underline{m} \neq -1}^{\bar{m}} \beta_m \cdot \text{Treated}_{im} \cdot d_{jm} + \mu_{ij} + \lambda_{jm} + \varepsilon_{ijm} \quad (1)$$

where dependent variable  $Y_{ijm}$  includes the indicator for delinquency, log delinquency amount, indicator for prepayment, and log prepayment amount of household  $i$  in stack  $j$  and on month  $m$ . The treatment dummy  $\text{Treated}_{im}$  indicates the treatment status for households.  $d_{jm}$  is a dummy variable that equals 1 if an observation in cohort  $j$  is  $m$  months away from the opening of the new subway station, with  $m = 0$  representing the month of opening. Positive values of  $m$  indicate months after the expansion, while negative values indicate months before.  $[\underline{m}, \bar{m}]$  refers to time horizons, where  $\underline{m} = -12$  and  $\bar{m} = 12$ . Our baseline specification controls for individual-by-stack fixed effect  $\mu_{ij}$  and month-by-stack fixed effect  $\lambda_{jm}$ .  $\varepsilon_{ijm}$  is the error term. Standard errors are clustered by postal code, the level at which treatment is assigned.

<sup>15</sup> As noted by [Baker et al. \(2022\)](#), the stacked DiD model is more efficient and easier to implement compared to alternatives such as those proposed by [Callaway and Sant'Anna \(2021\)](#) and [Sun and Abraham \(2021\)](#).

The month prior to the expansions,  $m = -1$ , serves as the reference point. Here, the coefficient  $\beta_m$  represents variance-weighted treatment effects across all cohorts. When the dependent variable is an indicator for delinquency or prepayment,  $m \geq 0$  ( $< 0$ ) reflects the probability change in delinquency or prepayment in month  $m$  after (before) subway expansions, relative to the month preceding the expansion. If the dependent variable is the logarithm of delinquency or prepayment amounts, the coefficient captures the monthly percentage change in these amounts after (before) the expansions. Thus, this can be interpreted as an event study analysis, testing for pre-trends when  $m < 0$ , and the persistence of effects when  $m \geq 0$ .

To estimate the average effects of subway expansions on delinquency and prepayment, we modify the dynamic specification as in Equation (1) to the following specification:

$$Y_{ijm} = \beta \cdot Treated_{im} \cdot Post_{jm} + \mu_{ij} + \lambda_{jm} + \varepsilon_{ijm} \quad (2)$$

where  $Post_{jm}$  is a dummy variable that equals 1 for all time periods after the opening of subway stations in cohort  $j$ . Here,  $\beta$  refers to the average changes of dependent variables as a result of subway expansions.

We start by analyzing the impact of subway expansions on delinquency rates. Column (1) of Table 3 reports the average effect of these expansions on the probability of households falling into monthly delinquency, as estimated from Equation (2). Based on our stacked Difference-in-Differences (DiD) estimation, the results show that subway expansions do reduce the monthly delinquency rate, with an average decrease of 4.42% (statistically significant at the 1% level). This indicates that improved access to public transit may contribute to reducing financial distress for households.

In order to validate the parallel-trends assumption underlying our stacked DiD specification, we examine the dynamic effect of subway expansions on the delinquency rate over a period spanning 12 months before and 12 months after the station opening. Panel A of Fig. 2 presents the month-by-month estimates, alongside the corresponding 95% confidence intervals based on the regression model outlined in Equation (1). We find that the coefficients on the lead indicators are generally small and statistically insignificant, lending support to the validity of the parallel trends assumption. In addition, we observe that the delinquency rate remains relatively stable before the station openings and does not begin to decrease until the third month after the expansions. This decline continues steadily over the first six months, reaching its lowest point around the seventh month, where it stabilizes at approximately 4%. The

**Table 3**  
Stacked DiD estimates for subway expansions – mortgage.

| Dependent Variable:          | Indicator for Delinquency | Log (Delinquency Amount) | Indicator for Prepayment | Log (Prepayment Amount) |
|------------------------------|---------------------------|--------------------------|--------------------------|-------------------------|
|                              | (1)                       | (2)                      | (3)                      | (4)                     |
| <i>Treated</i> × <i>Post</i> | −0.0442***<br>(0.0101)    | −0.392***<br>(0.0892)    | 0.0138**<br>(0.00701)    | 0.102<br>(0.0746)       |
| Observations                 | 2,484,192                 | 2,484,192                | 2,484,192                | 2,484,192               |
| $R^2$                        | 0.597                     | 0.603                    | 0.684                    | 0.737                   |
| Dep. Var. Mean               | 0.29                      | 7,212                    | 0.60                     | 84,420                  |
| Year-Month Fixed Effects     | Yes                       | Yes                      | Yes                      | Yes                     |
| Account Fixed Effects        | Yes                       | Yes                      | Yes                      | Yes                     |

*Notes:* Based on the mortgage repayment records in Delhi, this table reports the impact of the opening of a subway station between 2015 and 2019. The dependent variables encompass the indicator for delinquency, delinquency amount, the indicator for prepayment, and prepayment amount. The amounts are reported in their logarithm values. For the independent variable, *Treated* × *Post* an indicator variable that is equal to one after the opening of the subway station in the same postal code, and zero otherwise. Clustering is done at the postal code level. The robust standard errors are reported in parenthesis. Overall means of delinquency and prepayment amounts are reported in levels. \*, \*\*, and \*\*\* denote statistically significant levels at 10%, 5% and 1% respectively.

sustained reduction in delinquency rates suggest that there is a lasting impact of subway expansions, with the effect becoming particularly evident by the 12-month mark.

Next, we focus on the monthly delinquency amount. Column (2) of Table 3 shows that subway expansions leads to a 39.2% reduction in monthly delinquency amount (statistically significant at the 1% level). Panel B of Fig. 2 illustrates the dynamic effects of subway expansions on delinquency amounts. As before, there is no evidence of pre-treatment trends. The pattern of delinquency amount dynamics is similar to that of delinquency rate: it begins to decline three months after a station opens and continues to decrease over the next six months. The reduction reaches its lowest point, stabilizing at around 35%, between the sixth and ninth months. The impact of the subway expansions persists for more than 12 months.

We now turn to prepayment rates. Based on Column (3) of Table 3, we note that households' monthly prepayment rate increases by 1.38% (statistically significant at the 5% level) after the expansions on average. Panel C of Fig. 2 shows no evidence of pre-trends prior to treatment, indicating that treated and control groups followed similar trajectories before the subway expansions. Different from the response of delinquency, prepayment rate increases immediately on the month of subway expansions. The increments in prepayment rate last for three months. After a dip on the fourth month, the positive effect of the expansions holds significant until 6th month. Interestingly, the immediate increase in prepayment rates suggests that households may prioritize reducing their debt burdens in response to improved transportation access. This shift could reflect a broader financial adjustment, as households allocate more resources toward repaying existing loans following subway expansion.

Finally, we analyze the monthly prepayment amount. Column (4) of Table 3 reports a 10.2% increase in prepayment amounts following subway expansions, though this effect is not statistically significant at the 10% level. Panel D of Fig. 2 shows that the dynamic effects of prepayment amounts closely mirror those of prepayment rates: there is a clear increase in the immediate post-expansion period that fades over time. The difference in statistical significance arises because the rate indicator captures the extensive margin—whether households make a prepayment at all. In contrast, the amount measure captures the intensive margin, which is more variable and subject to greater noise. As a result, we find statistically significant increases in prepayment amounts during the first three months after the expansions, but the transitory nature of these effects combined with higher variance in amounts leads to a statistically insignificant aggregate effect.

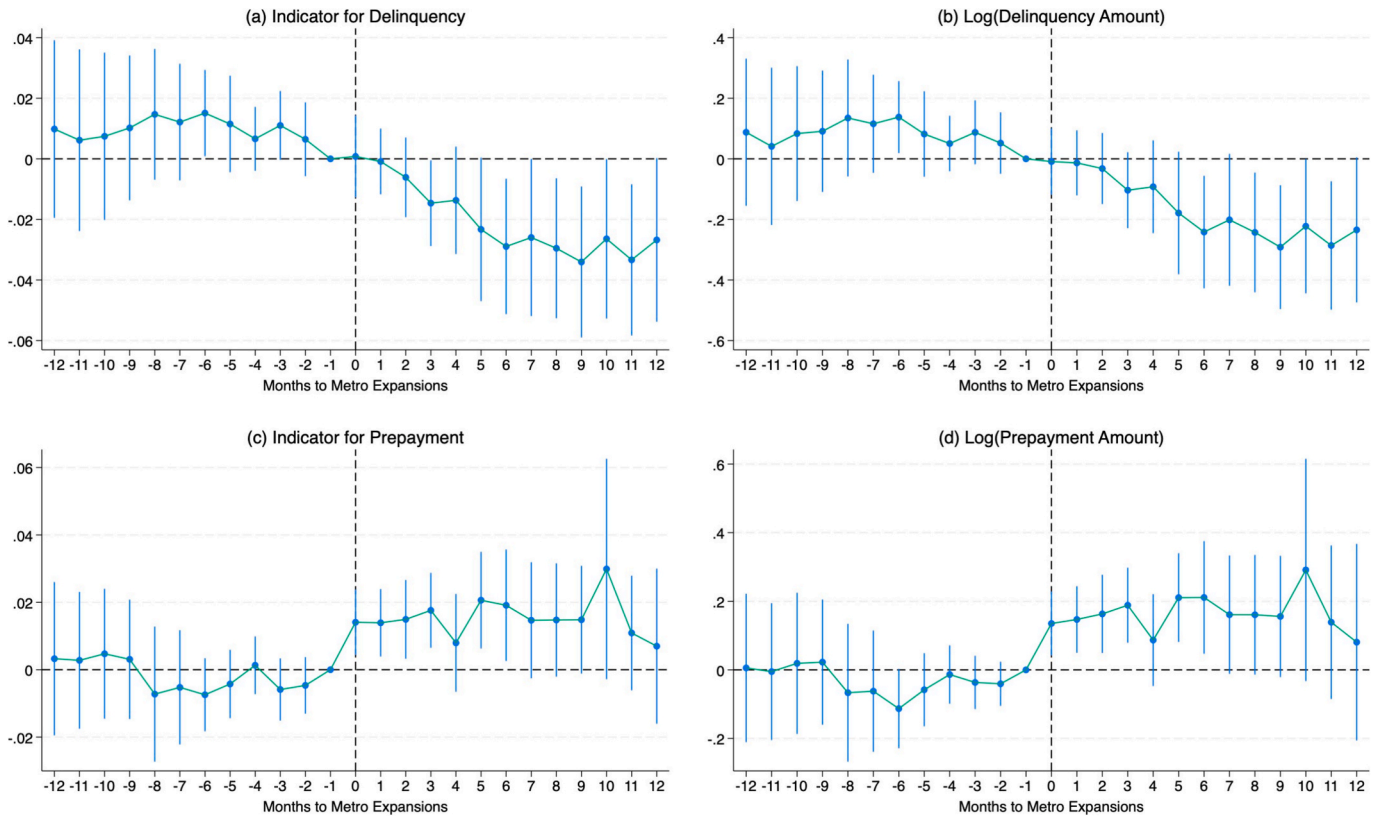
In all, our findings suggest that subway expansions reduce households' monthly delinquency rates, lower delinquency amounts, and lead to an increase in monthly prepayment rates, even if the effect on prepayment amounts is more short-lived.

### 3.2. Different control groups by distance

We next explore how the effects of subway expansions vary with distance from the nearest subway station. Our hypothesis is that the positive effects decrease as distance increases. While our baseline analysis centers on households within the same postal code as the new station, proximity likely matters: households located very close to a station should experience stronger effects than those farther away. In addition, subway stations situated near postal code boundaries may create spillover effects, as some households in adjacent postal codes may be closer to the station than those living within the designated treatment area.

To test for these distance-based differences, we re-estimate our main results using two alternative sets of control groups, while keeping the treatment group unchanged. In the first specification, the control group consists of postal code areas without subway expansions during the sample period but geographically adjacent to treated areas, capturing 48 neighboring postal codes. In the second specification, the control group





**Fig. 2.** Event study of subway expansions. *Notes:* Panel (a), (b), (c), and (d) present the dynamic effects of subway expansions on indicator for delinquency, logarithmic delinquency amount, indicator for prepayment, and logarithmic prepayment amount respectively based on the baseline specification. Positive  $m$  refers to  $m_{th}$  month after expansions and negative  $m$  refers to  $m_{th}$  month before expansions.  $m = 0$  refers to the month of subway expansions. We choose one month before expansions  $m = -1$  as our base period. Vertical lines on dots denote 95 percent confidence interval.

is drawn from the 11 remaining postal codes without subway expansions that are located farther away, removing the neighboring areas. This approach is similar to the “donut” design used in existing studies (Barreca et al., 2011), which excludes observations located very close to treated units. Table 4 reports the re-estimated average effects based on these alternative control groups.

We first re-estimate the results using neighboring postal code areas as the control group. As shown in Table 4, Panel A, the delinquency rate falls by 3.97% (statistically significant at the 1% level) and the delinquency amount declines by 33.9% (statistically significant at the 1% level) after subway expansions. These estimates are smaller in magnitude than the baseline results, and the effects on prepayment rates and amounts are no longer statistically significant. This attenuation is consistent with the idea that households in adjacent postal code areas also benefit from improved transit access. As a result, the difference between treated and control households narrows when neighboring areas are used as the control group.

In Table 4, Panel B, we next use postal code areas located farther from subway stations as the control group. Relative to these households, those living in the same postal code as the new stations experience a 7.95% reduction in delinquency rates (statistically significant at the 1% level) and an 80.8% decline in delinquency amounts (statistically significant at the 1% level). In addition, the prepayment rate increases by 3.59% (statistically significant at the 1% level) and the prepayment amount rises by 31.5% (statistically significant at the 1% level) following subway expansions. These effects are larger than the baseline estimates, highlighting the importance of proximity to subway stations in shaping mortgage outcomes. The comparison reinforces our interpretation that the benefits of subway access diminish with distance and that households located closer to the new stations experience the largest improvements in mortgage performance. To complement these results,

we explicitly quantify spillover effects by treating adjacent areas as “treated” and using more distant areas as the control group. This approach allows us to directly measure how subway expansions affect households just outside the treated postal codes and to gauge the extent of these spillover benefits. Details of this exercise are provided in Online Appendix C.

#### 4. Role of automobile purchases

Having established that the opening of a subway station affects both mortgage delinquency and prepayment decisions, we now investigate the underlying mechanism. Here, we focus on the role of reduced transportation costs, which can free up resources that households use to prepay mortgages or lower their delinquency risk. To test this channel directly, we analyze automobile purchases, one of the largest discretionary expenses for many households and therefore a strong proxy for financial capacity. Examining changes in vehicle-buying behavior allows us to assess the extent to which households substitute private vehicle ownership with subway use. We then draw on income information from the mortgage dataset to evaluate whether these responses vary across income groups, particularly between lower- and higher-income households. Together, these analyses provide a clearer picture of how subway expansions reduce vehicle expenditures, ease financial constraints, and ultimately improve mortgage performance.

##### 4.1. Effects of subway expansions on automobile purchases

Why do households become less delinquent and increase prepayments when a subway station opens nearby? A plausible explanation is that financially constrained households substitute away from private vehicle use toward the more affordable subway system, generating

**Table 4**  
Effects of subway expansions on mortgage – restricted control groups.

| Panel A: Close Postal Code Areas |                           |                          |                          |                         |
|----------------------------------|---------------------------|--------------------------|--------------------------|-------------------------|
| Dependent Variable:              | Indicator for Delinquency | Log (Delinquency Amount) | Indicator for Prepayment | Log (Prepayment Amount) |
|                                  | (1)                       | (2)                      | (3)                      | (4)                     |
| <i>Treated</i> × <i>Post</i>     | −0.0397***<br>(0.0109)    | −0.339***<br>(0.0963)    | 0.0109<br>(0.00758)      | 0.0726<br>(0.0776)      |
| Observations                     | 2,202,079                 | 2,202,079                | 2,202,079                | 2,202,079               |
| R <sup>2</sup>                   | 0.599                     | 0.607                    | 0.681                    | 0.736                   |
| Dep. Var. Mean                   | 0.29                      | 7,412                    | 0.60                     | 86,445                  |
| Year-Month Fixed Effects         | Yes                       | Yes                      | Yes                      | Yes                     |
| Account Fixed Effects            | Yes                       | Yes                      | Yes                      | Yes                     |
| Panel B: Far Postal Code Areas   |                           |                          |                          |                         |
| Dependent Variable:              | Indicator for Delinquency | Log (Delinquency Amount) | Indicator for Prepayment | Log (Prepayment Amount) |
|                                  | (1)                       | (2)                      | (3)                      | (4)                     |
| <i>Treated</i> × <i>Post</i>     | −0.0795***<br>(0.0105)    | −0.808***<br>(0.0952)    | 0.0359***<br>(0.00633)   | 0.315***<br>(0.0718)    |
| Observations                     | 466,657                   | 466,657                  | 466,657                  | 466,657                 |
| R <sup>2</sup>                   | 0.592                     | 0.590                    | 0.698                    | 0.742                   |
| Dep. Var. Mean                   | 0.28                      | 8,283                    | 0.62                     | 94,096                  |
| Year-Month Fixed Effects         | Yes                       | Yes                      | Yes                      | Yes                     |
| Account Fixed Effects            | Yes                       | Yes                      | Yes                      | Yes                     |

*Notes:* This table reports the impact of the opening of a subway station between 2015 and 2019, based on two different control groups. Panel A (B) presents the results using the postal code areas close to (far from) the treatment areas as the control group. The dependent variables encompass the indicator for delinquency, delinquency amount, the indicator for prepayment, and prepayment amount. The amounts are reported in their logarithm values. For the independent variable, *Treated* × *Post* an indicator variable that is equal to one after the opening of the subway station in the same postal code, and zero otherwise. Clustering is done at the postal code level. The robust standard errors are reported in parentheses. Overall means of delinquency and prepayment amounts are reported in levels. \*, \*\* and \*\*\* denote statistically significant levels at 10%, 5% and 1% respectively.

savings on transportation costs. By cutting down on expenses associated with car ownership, such as loan payments, fuel, maintenance, and insurance, households can improve their financial position. These savings ease financial constraints, enabling borrowers to better manage mortgage payments and, in some cases, make early prepayments.

In what follows, we analyze how subway expansions affect vehicle purchases and spending. Using nearly 3 million vehicle registration records aggregated at the postal code-by-month level, we estimate stacked DiD models following Equation (2). Instead of using prepayment and delinquency as the dependent variables, we replace them with the number of vehicles purchased and total vehicle spending. This analysis allows us to directly observe whether improved transit access changes household transportation choices, which is a key mechanism underlying the mortgage repayment results. If households are substituting away from private vehicle ownership and the associated costs, we would expect to see fewer vehicle purchases and lower overall vehicle spending in areas receiving new subway stations.

There is strong evidence that subway access influences household

vehicle-buying behavior. Table 5 Panel A shows that subway expansions reduce demand for four-wheelers, while the effects on two- and three-wheelers are smaller and statistically insignificant. The number of four-wheelers sold falls by approximately 7 vehicles per month (statistically significant at the 10% level), while sales of two- and three-wheelers decline by about 2 units, though these reductions are not statistically significant at the 10% level. More importantly, total vehicle expenditures fall by 6.52% in areas with new subway stations (statistically significant at the 5% level), and the average spending per vehicle decreases by 4.69% (statistically significant at the 1% level). Our findings thus suggest that subway expansions not only shift preferences away from four-wheelers but also reduce overall vehicle spending.

Similar to our analysis of mortgage repayment, we then test the sensitivity of these results to the choice of control group. Table A.1, Panel A uses neighboring postal codes as the control group and shows no statistically significant effects of subway expansions on vehicle purchases. Panel B instead uses distant postal codes and yields much stronger results. Following subway openings, the number of registered

**Table 5**  
Stacked DiD estimates for subway expansions – vehicle.

| Panel A: Vehicles Purchase           |                                      |                                     |                            |                           |
|--------------------------------------|--------------------------------------|-------------------------------------|----------------------------|---------------------------|
| Dependent Variable:                  | Two- and Three-Wheeler               | Four-Wheeler                        | Average Spending           | Total Spending            |
|                                      | (1)                                  | (2)                                 | (3)                        | (4)                       |
| <i>Treated</i> × <i>Post</i>         | −2.236<br>(9.560)                    | −6.956*<br>(3.673)                  | −0.0469***<br>(0.0133)     | −0.0652**<br>(0.0256)     |
| Observations                         | 45,358                               | 45,358                              | 45,358                     | 45,358                    |
| R <sup>2</sup>                       | 0.922                                | 0.924                               | 0.418                      | 0.889                     |
| Dep. Var. Mean                       | 415                                  | 143                                 | 13                         | 18                        |
| Year-Month Fixed Effects             | Yes                                  | Yes                                 | Yes                        | Yes                       |
| Postal Code Fixed Effects            | Yes                                  | Yes                                 | Yes                        | Yes                       |
| Panel B: Vehicles Purchase (by type) |                                      |                                     |                            |                           |
| Dependent Variable:                  | Two- and Three-Wheeler: High Quality | Two- and Three-Wheeler: Low Quality | Four-Wheeler: High Quality | Four-Wheeler: Low Quality |
|                                      | (1)                                  | (2)                                 | (3)                        | (4)                       |
| <i>Treated</i> × <i>Post</i>         | 4.729<br>(3.691)                     | −6.965<br>(7.869)                   | 2.079*<br>(1.196)          | −9.034**<br>(3.861)       |
| Observations                         | 45,358                               | 45,358                              | 45,358                     | 45,358                    |
| R <sup>2</sup>                       | 0.860                                | 0.914                               | 0.910                      | 0.903                     |
| Dep. Var. Mean                       | 85                                   | 330                                 | 41                         | 102                       |
| Year-Month Fixed Effects             | Yes                                  | Yes                                 | Yes                        | Yes                       |
| Postal Code Fixed Effects            | Yes                                  | Yes                                 | Yes                        | Yes                       |

*Notes:* Based on data aggregated at the postal code level, this table reports the impact of the opening of a subway station on the purchase on vehicles within the same postal code area. The treatment indicator is equal to one after the opening of the subway station, and zero otherwise. In Panel A, the dependent variables include the absolute number of registered two- and three-wheelers, the absolute number of registered four-wheelers, the average costs of each vehicle, the total spending on vehicles at postal code level. In Panel B, the dependent variables relate to the quality of four-wheelers and two- and three-wheelers respectively. We consider a vehicle to be high (low) quality if its price is higher (lower) than the average price of the vehicle registered during the time period. Robust standard errors are in parentheses, clustered at postal code level. Overall means of average spending and total spending are reported in log form. \*, \*\* and \*\*\* denote significance level at 10%, 5% and 1% respectively.

four-wheelers declines by about 15 units (statistically significant at the 1% level), while average vehicle-related spending and total monthly spending fall by 28.5% and 36.3%, respectively (both statistically significant at the 1% level). These spatial patterns mirror those observed in the mortgage outcomes and reinforce the vehicle-substitution channel: subway expansions reduce reliance on private vehicles, easing household liquidity constraints and improving mortgage performance.

To better understand which households drive the reduction in vehicle expenditures, we examine heterogeneity in the response of vehicle purchases to subway expansions by distinguishing between high-quality and low-quality vehicles. This approach is necessary because the dataset lacks direct measures of household income, making vehicle quality a plausible proxy for financial capacity. Households with greater financial resources are generally more willing and able to purchase higher-quality and, more expensive vehicles, while financially constrained households tend to purchase lower-quality vehicles. Analyzing heterogeneity by vehicle quality therefore helps identify which groups are most affected by improved access to public transportation.

As mentioned earlier, high-quality vehicles are defined as those priced above the average within each vehicle category, while low-quality vehicles are priced below the average. Table 5, Panel B shows that the reduction in four-wheeler registrations is concentrated entirely among low-quality vehicles. After a subway station opens, registrations of low-quality four-wheelers fall by 9 vehicles (statistically significant at the 5% level). In contrast, registrations of high-quality four-wheelers increase modestly by 2 vehicles (statistically significant at the 10% level). Our findings indicate that households purchasing low-quality vehicles are the primary drivers of the observed reduction in vehicle expenditures. Households purchasing high-quality vehicles appear less likely to substitute away from private car use, suggesting that the financial savings from switching to public transit are less meaningful for their budgets.

Online Appendix D provides additional evidence on automobile purchases and substitution across vehicle types to complement these main results. Specifically, we examine households' likelihood of substituting between four-wheelers and other vehicle types using the same stacked DiD framework. Consistent with the main findings, we find that the probability that a newly registered vehicle is a four-wheeler decreases by 0.39% following subway openings. We also document a 1.2% decline in the market share of four-wheelers. These results suggest that households increasingly substitute away from four-wheelers toward two- and three-wheelers, which are generally cheaper and more compatible with using the subway for longer trips. This substitution further reinforces the mechanism that improved transit access reduces vehicle-related expenditures, especially among households that are most financially constrained.

In sum, subway expansions lead to sizable reductions in vehicle expenditures, particularly among financially constrained households. We next quantify how large these savings are relative to typical automobile loan obligations using institutional evidence and back-of-the-envelope calculations.

#### 4.2. Automobile loans and back-of-the-envelope calculations

To evaluate how reductions in vehicle expenditures could affect household budgets, we draw on automobile loan conditions in Delhi and conduct simple back-of-the-envelope calculation. As one of the largest discretionary expenses, automobile purchases are commonly financed through readily accessible loans. This makes auto loan terms a useful benchmark for estimating the potential savings from foregone car purchases.

Between 2015 and 2020, the average price of a new car in Delhi was approximately ₹800,000 (about USD 11,000 based on the 2020 exchange rate). Car financing was widely available and typically offered on favorable terms. For instance, under the State Bank of India (SBI)'s

New Car Loan Scheme, salaried individuals could borrow up to 48 times their net monthly income, while existing SBI home loan customers enjoyed even more favorable conditions, including no loan amount caps and lower minimum income requirements. Downpayments were typically modest at around 10% of the vehicle's on-road price, and some lenders such as HDFC Bank even offered zero-downpayment options.<sup>16</sup> Car loans generally carried interest rates near 10% and repayment tenures of one to seven years. For a standard five-year loan with a 10% downpayment and a 10% interest rate, the Equated Monthly Installment (EMI) on a typical ₹800,000 car purchase would be approximately ₹15,000.

Using these institutional details, we benchmark the improvement in household repayment capacity implied by our main results. We find that subway expansions lead to a 39.2% decline in delinquency and a 10.2% increase in prepayment, which together translate into an average improvement in repayment capacity of ₹11,438.<sup>17</sup> To put this into perspective, this amount is roughly 75% of the estimated ₹15,000 monthly EMI on a standard car loan. Thus, the savings associated with reduced automobile expenditures are large enough to nearly cover the majority of a typical monthly car loan payment, highlighting the magnitude of the liquidity relief generated by subway expansions.<sup>f</sup>

Although this amount may be less consequential for wealthier households, it is substantial for lower- and middle-income households with tighter financial constraints. For these households, the ability to redirect savings from reduced automobile expenditures can make the difference between falling behind on mortgage payments and staying current. This benchmark reinforces the earlier heterogeneity results, which showed that the largest effects of subway expansions were concentrated among purchases of low-quality vehicles, typically made by lower- and middle-income households. In Online Appendix E, we further examine automobile affordability for households in our sample by showing that car ownership was financially feasible but often left households heavily leveraged. Thus, avoiding car purchases can substantially ease liquidity constraints.

To directly validate this mechanism, we return to the mortgage dataset and examine whether the effects of subway expansions differ across income groups.

#### 4.3. Additional evidence of income

The mortgage dataset provides detailed information on borrowers' income at the time of loan origination, which enables us to test whether subway expansion effects differ across income groups. This heterogeneity analysis complements the patterns observed in vehicle purchases, where buyers of lower-quality vehicles, who are more likely to be lower-income households, exhibited the largest reductions in automobile expenditures.

Formally, we estimate the following specification:

$$Y_{ijm} = \beta \cdot \text{Treated}_{im} \cdot \text{Post}_{jm} + \sum_{k \in K} \beta_k \cdot \text{Treated}_{im} \cdot \text{Post}_{jm} \cdot I_{jk} + \mu_{ij} + \lambda_{jm} + \varepsilon_{ijm} \quad (3)$$

where  $K$  is a leave-one-out set of mortgage applicant characteristics. Categorical variable  $I_{jk}$  equals to 1 if a applicant is in a group with characteristic  $k$ . The base category is the characteristic being dropped. Then each  $\beta_k$  can be interpreted as the differences in the effects of metro expansions on mortgage delinquency and prepayment, compared applicants with characteristic  $k$  to those with corresponding base category.

Our hypothesis is that lower-income households are more likely to switch from private vehicles to subway transportation, which may

<sup>16</sup> For more details, see "SBI Car Loan Eligibility Table" and "HDFC Bank Xpress Car Loan". Major banks like ICICI and Axis Bank provide comparable products with similar eligibility and repayment terms.

<sup>17</sup> This amount is calculated as the sum of 39.2% of the average delinquency amount (₹7,212) and 10.2% of the average prepayment amount (₹84,420).

improve their financial flexibility. However, some low-income households may not be able to afford a vehicle at all, limiting the benefits they gain from subway expansions. To account for this variation, we divide households into three groups: lowest-income (below the 25th percentile), lower-middle-income (between the 25th and 50th percentiles), and above-median (above the 50th percentile). This approach helps us better identify differences in financial behavior across income levels. With reference to Equation (3), the above-median group serves as the omitted category for comparison. We present the results in Table 6.

Consistent with our findings in the vehicles market, we observe that lower-middle-income households experience a 5.86% larger decrease in delinquency rates (statistically significant at the 1% level) and a 52.5% greater reduction in delinquency amounts (statistically significant at the 1% level) compared to households with income above the median. Additionally, the prepayment rate and amount for lower-middle-income households increase by 3.7% (statistically significant at the 10% level) and 37.8% (statistically significant at the 5% level), respectively, more than their above-median counterparts following subway expansions. In contrast, the effects of subway expansions on delinquency and prepayment for lowest-income households are not significantly different from those observed in households with income above the median. Our findings highlight the key role of automobile expenditures in mortgage repayments. Lower-middle-income households are more likely to replace private vehicle use with subway transportation, enhancing their financial flexibility. However, lowest-income households, who likely could not afford private vehicles prior to the subway expansions, do not benefit from reduced vehicle expenditures.

Previous studies often classify households into broad high- and low-income groups, implicitly treating welfare effects of commuting improvements as similar within these categories (Balboni et al., 2020; Lee and Tan, 2024). By focusing on automobile purchases, we seek to distinguish between the lowest-income and lower-middle-income

households, providing a more differentiated view of how subway expansions affect different segments of the income distribution. Our findings indicate that subway expansions reduce financial pressures for households with constrained but not minimal resources, offering clearer guidance for targeted transportation policies.

## 5. Discussion

We now consider alternative explanations, assess the robustness of our findings, and examine their policy relevance. First, we explore other potential mechanisms, such as increases in earnings or local economic activity, that could account for the observed improvements in mortgage performance following subway expansions. Second, we test the robustness of our results using a series of empirical checks, including falsification tests and alternative model specifications. These checks consistently support the validity of our main findings. Finally, we discuss the broader policy implications, focusing on how transit infrastructure can improve household financial resilience.

### 5.1. Alternative explanations

We begin by evaluating alternative channels through which subway expansions may influence household mortgage behavior, beyond the reduction in automobile expenditures documented earlier. Two mechanisms warrant particular attention. The first is that improved access to public transit raises household income, either by expanding employment opportunities or increasing individual productivity. The second is that subway expansions spur localized economic development, improving neighborhood conditions and property values. However, as we demonstrate below, the empirical evidence provides limited support for either mechanism. We examine each in turn.

#### 5.1.1. Individual income

One potential mechanism is that subway expansions increase household income by enhancing labor productivity or enabling greater workforce participation. Reduced commuting time may improve job performance and, over time, translate into higher wages or bonuses. To evaluate this channel, we draw on borrower-level and aggregated income data, along with a series of heterogeneity tests. While we lack detailed labor market information at the postal code level, the rich borrower demographics in our dataset allow us to indirectly assess this mechanism.

As a first test of the productivity-based income channel, we compare the effects of subway expansions on borrowers employed in the public sector versus those in the private sector. The rationale is that private sector compensation tends to be more closely tied to performance and is therefore more likely to respond to productivity gains from reduced commuting time. If this mechanism explains the results, private sector borrowers should experience larger improvements in mortgage outcomes, such as lower delinquency rates or higher prepayment rates, compared to public sector borrowers. Borrowers are classified based on their reported occupation at the time of mortgage application. Public sector employment includes positions in central government services, defense establishments, public sector undertakings, and state government services. We re-estimate our benchmark regression separately for each group. As shown in Table A.2, Panel A, the effects of subway expansions do not differ significantly across sectors. This lack of heterogeneity suggests that the productivity or bonus channel is unlikely to account for the observed improvements in mortgage performance.

Building on the previous analysis, we next consider whether subway expansions improve household finances by increasing labor supply or participation rates. Shorter commutes and lower transportation costs may reduce barriers to labor market entry, particularly for individuals facing time constraints (Gutiérrez-i Puigarnau and van Ommeren, 2010). Greater labor force participation would likely raise household income and, in turn, improve mortgage performance. To evaluate this

**Table 6**  
Heterogeneous effects based on income.

| Dependent variable:                                                    | Indicator for Delinquency | Log (Delinquency Amount) | Indicator for Prepayment | Log (Prepayment Amount) |
|------------------------------------------------------------------------|---------------------------|--------------------------|--------------------------|-------------------------|
|                                                                        | (1)                       | (2)                      | (3)                      | (4)                     |
| <i>Treated</i> × <i>Post</i>                                           | −0.0404**<br>(0.0180)     | −0.325*<br>(0.168)       | 0.00986<br>(0.0116)      | 0.0366<br>(0.123)       |
| <i>Treated</i> × <i>Post</i> × 1 (Annual Income ≤ ₹100,000)            | −0.0128<br>(0.0242)       | −0.110<br>(0.193)        | −0.00154<br>(0.0231)     | 0.118<br>(0.267)        |
| <i>Treated</i> × <i>Post</i> × 1 (₹100,000 < Annual Income ≤ ₹400,000) | −0.0586***<br>(0.0200)    | −0.525***<br>(0.180)     | 0.0370*<br>(0.0192)      | 0.378**<br>(0.179)      |
| Observations                                                           | 1,101,831                 | 1,101,831                | 1,101,831                | 1,101,831               |
| R <sup>2</sup>                                                         | 0.584                     | 0.593                    | 0.681                    | 0.733                   |
| Dep. Var. Mean                                                         | 0.29                      | 7,212                    | 0.60                     | 84,420                  |
| Year-Month Fixed Effects                                               | Yes                       | Yes                      | Yes                      | Yes                     |
| Account Fixed Effects                                                  | Yes                       | Yes                      | Yes                      | Yes                     |

*Notes:* This table reports the heterogeneous effects of the opening of subway stations by household's annual income before subway expansions. The treatment indicator is equal to one after the opening of the subway station, and zero otherwise. ₹100,000 and ₹400,000 are the 25th and 50th percentile of annual income in our dataset. The base category is the households whose income is above the median. Robust standard errors are in parentheses, clustered at postal code level. Overall means of delinquency and prepayment amounts are reported in levels. \*, \*\* and \*\*\* denote significance level at 10%, 5% and 1% respectively.



mechanism, we examine heterogeneous effects by gender.<sup>18</sup> If the labor supply channel is relevant, women who often face tighter time constraints due to commuting and household responsibilities, may be more likely to join the workforce or increase their working hours when commuting becomes easier (Liu et al., 2025). These changes could lead to lower delinquency risk or a higher likelihood of prepayment. However, as shown in Panel B of Table A.2, we find no significant differences in the effects of subway expansions on mortgage outcomes by gender. This lack of heterogeneity suggests that increased labor supply, particularly among women, is unlikely to be the primary mechanism behind the observed effects.

We further assess the job access mechanism by classifying borrowers according to the location of their purchased properties, distinguishing between city-center and suburban areas. This classification is motivated by the expectation that suburban residents typically face longer and more expensive commutes and therefore stand to gain more from reductions in travel time and transportation costs resulting from subway expansions. If improved job access were a key driver of mortgage performance, we would expect to see stronger treatment effects among suburban borrowers, who benefit more at the margin from enhanced transit connectivity. However, as shown in Panel C of Table A.2, the treatment effects do not differ significantly between borrowers in suburban and central locations. This lack of geographic heterogeneity, combined with the earlier null findings by gender, suggests that improvements in job access are unlikely to explain the observed changes in mortgage delinquency and prepayment behavior.

Finally, we test whether subway expansions directly affect borrower income. A key limitation of our dataset is that income is recorded only once, at the time of mortgage origination, which prevents us from observing individual income changes over time. As a result, we cannot apply a standard Difference-in-Differences strategy at the individual level. To address this constraint, we construct a panel at the postal code and month level by calculating the average income of all households initiating mortgages in each location and period. This approach allows us to trace income trends across neighborhoods before and after subway openings, including for new entrants outside our main sample.

One potential concern with this strategy is that improved transit access may alter the composition of borrowers, either by attracting lower-income households seeking greater liquidity or by drawing higher-income buyers anticipating capital gains. To test for such selection, we compare observable borrower characteristics before and after station openings, including age, gender, and private sector employment share. As shown in Table A.3, we find no statistically significant differences. Having established that the borrower pool remains comparable, we proceed to examine income trends using the postal code-level panel. This specification includes borrowers who purchased after the expansions, allowing us to capture post-treatment income levels. Columns (1) and (2) of Table A.4 report a small and statistically insignificant decline in average income in the months following subway openings. This result undermines the income-based explanation and aligns with existing evidence suggesting that any income gains from subway expansions tend to materialize gradually. For example, Koh et al. (2025) find that subway expansions increase inventor productivity only after five years.

### 5.1.2. Macroeconomic conditions

Another potential channel is that subway expansions improve local macroeconomic conditions, which in turn enhance mortgage performance. Greater accessibility may attract investment, increase housing demand, and spur neighborhood development. These effects could raise

<sup>18</sup> We acknowledge that, although mortgage and repayment decisions reflect household-level dynamics, the mortgage applicant is typically the household head and primary financial decision-maker. The recorded gender in our dataset thus serves as a meaningful, though imperfect, proxy for the individual responsible for managing the mortgage.

property values, increase household equity, and reduce delinquency or increase prepayment rates.

To assess this hypothesis, we first use data on property values and loan amounts around subway openings. Using the same postal code and month panel structure as in the income analysis, we estimate a stacked Difference-in-Differences model to track local housing market dynamics. Columns (3) to (8) of Table A.4 report the results. Although some observations are excluded due to missing loan and income information,<sup>19</sup> we find no statistically or economically significant changes in either property values or loan sizes. While we acknowledge that sample selection remains a potential concern, we have shown earlier in Table A.3 that households purchasing before and after subway expansions are similar in age, gender and private sector employment. This supports the comparability of the samples and further weakens the case for a price-based explanation.

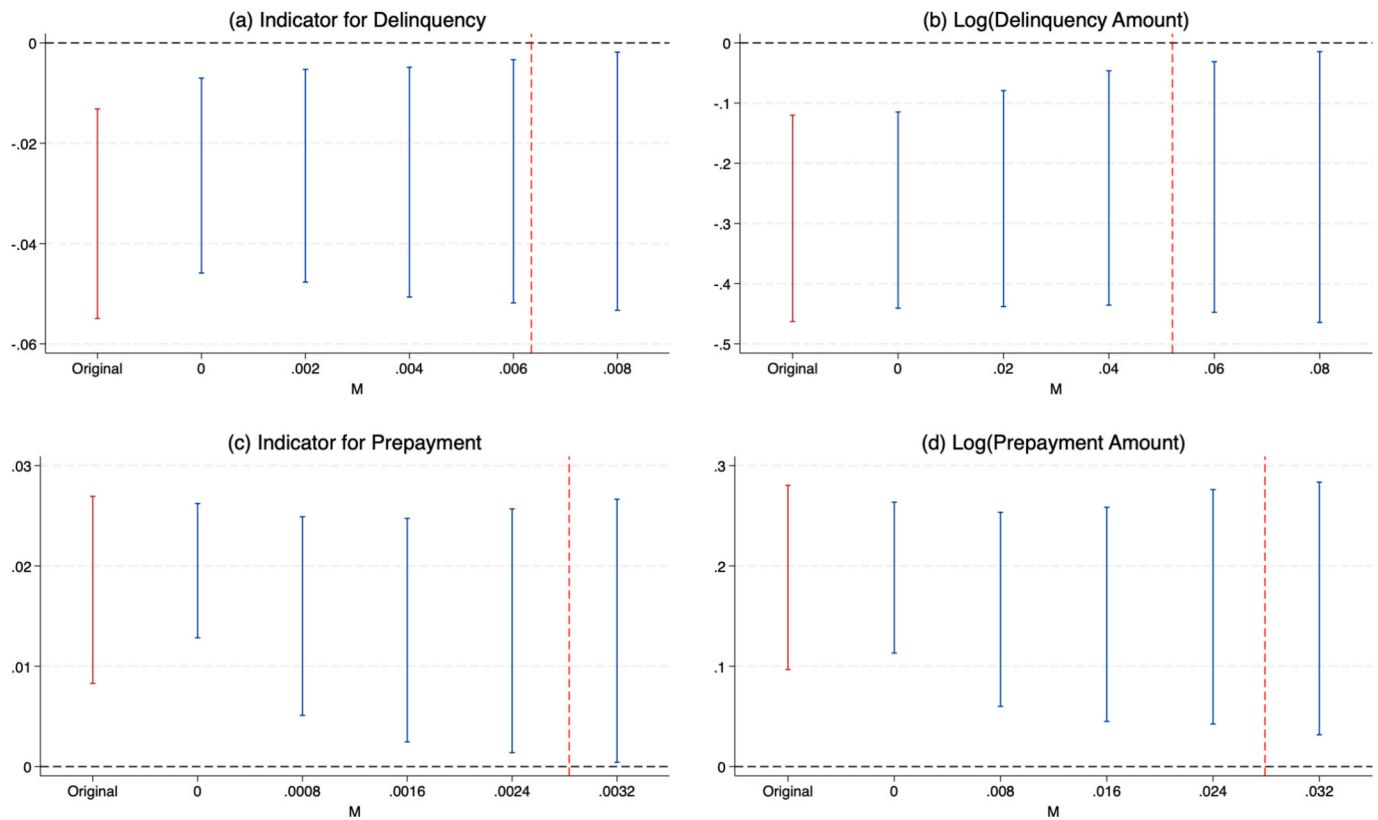
We further evaluate the housing wealth channel by testing for heterogeneity based on loan-to-value (LTV) ratios. Rising property prices reduce the risk of negative equity, which occurs when a home's value falls below the outstanding mortgage balance. This reduces the likelihood of strategic default. If housing appreciation were driving improvements in mortgage performance, we would expect stronger effects among high-LTV borrowers, who face greater downside risk. Instead, Table A.5 shows the opposite pattern: delinquency rates increased by 4.33 percent for high-LTV households relative to low-LTV households after subway openings, a difference that is statistically significant at the 10 percent level. This finding contradicts the housing wealth explanation. Instead, it aligns with recent research indicating that subway expansions have limited effects on local productivity or amenities (Severen, 2023). Moreover, housing markets are typically forward-looking, with price effects concentrated around project announcements rather than completion (Gupta et al., 2022), suggesting that improvements in mortgage performance are unlikely to be driven by housing wealth effects.

A related concern is that official transaction prices may understate true property values due to unreported "black money" payments (e.g., Anagol et al., 2023; Agarwal et al., 2020). If subway expansions increased property values primarily through informal payments, these effects might not appear in official records. However, we note that it is unlikely to bias our findings. First, by focusing on station openings rather than announcements, we reduce the risk of capturing anticipatory price adjustments in either formal or informal markets. Second, informal payments typically move in proportion to declared prices. The absence of a change in official prices following subway openings suggests that informal values likely remained unchanged as well.

To further address this concern, we draw on subjective housing price expectations, which provide an indirect proxy for underlying valuation dynamics. Specifically, we use household expectation data from the Reserve Bank of India's Households' Inflation Expectations Survey, covering March 2015 to January 2020. The survey asks Delhi residents whether they expect housing prices to rise, fall, or remain unchanged over three-month and one-year horizons. Prior work has shown that individuals' housing price expectations are shaped by personal experience and local context (Kuchler and Zafar, 2019), making this approach appropriate. Using the micro-level data, we match respondents to their postal codes and construct an indicator for positive price expectations. Applying the same stacked Difference-in-Differences framework, we find no change in the share of respondents expecting price increases following subway openings (Table A.6). Since these beliefs likely reflect both formal and informal market signals, the absence of a shift provides additional support that unreported "black money" transactions are not confounding our results.

Taken together, the evidence provides little support for income- or

<sup>19</sup> Observations for income and loan amount are fewer than for housing value because of missing entries in the mortgage data.



**Fig. 3.** Diagnose of pre-trend – Rambachan and Roth (2023). *Notes:* This figure presents the sensitivity analysis following Rambachan and Roth (2023). Panels (a) and (b) report estimated 90% confidence intervals for different values of  $M$ , the sensitivity parameter governing the size of allowable deviations from parallel trends, for the coefficients on  $Treated_i \times Post_0$ . Panels (c) and (d) report the corresponding intervals for the coefficients on  $Treated_i \times Post_3$ . The red dotted lines indicate the benchmark of half a standard error, as used by Mehic (2023) and Ge et al. (2025). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

macro-based explanations of the observed improvements in mortgage performance. Across multiple tests, we find no significant differences in treatment effects by occupation, gender, or location, nor any indication that average income levels rose following subway expansions. Similarly, we observe no meaningful changes in property values, loan amounts, or expectations of future housing prices. These findings suggest that neither enhanced job access nor broader neighborhood economic gains are driving the results. Instead, the evidence points toward a more direct mechanism: the reduction in commuting costs, which immediately improves household liquidity and may alleviate financial stress, thereby lowering delinquency and increasing prepayment.

## 5.2. Robustness

To assess the credibility of our identification strategy, we implement a set of robustness checks. We first consider anticipatory behavior and violations of the parallel trends assumption, and then examine possible confounding factors that could bias the estimates.

**Anticipation and Attention.** A key assumption of our Difference-in-Differences framework is that households did not adjust their mortgage behavior in anticipation of subway station openings. This assumption is plausible because, although the expansion plan was announced as early as 2011, most stations became operational only between 2015 and 2018. This long implementation lag, combined with limited updates or interim project milestones, reduces the likelihood that households could have precisely timed their financial decisions based on future transit access. Infrastructure delays are also common in the Indian context, further undermining the credibility of station openings as predictable events. In this setting, it is unlikely that households would have made meaningful adjustments to long-term mortgage behavior based solely on early

announcements.

We formally test this assumption using two placebo exercises. First, we assign a pseudo-treatment date three years prior to the actual opening and re-estimate our baseline model. As shown in Table A.7, Panel A, we find no significant effects on mortgage outcomes, indicating that long-term anticipatory behavior is unlikely to be a major driver of our results.<sup>20</sup> Second, we test for short-term anticipation by assigning treatment six months before station opening, a period when construction activity is typically visible and might begin to influence household decisions. Nonetheless, Table A.7, Panel B, shows no significant change in either delinquency or prepayment outcomes leading up to the opening, reinforcing the conclusion that households primarily responded to actual station access.

Behavioral evidence from Google Trends provides further support. Fig. A.1 shows that public search interest in “Delhi Metro” spiked around the 2011 announcement but declined steadily throughout the construction phase. Interest only rose again in the months surrounding the most intensive wave of station openings. This pattern suggests that while information about the project was publicly available, household attention and likely behavioral responses were more closely aligned with realized access rather than early announcements.

**Parallel Trends and Sensitivity.** Beyond anticipatory behavior, we formally examine the validity of the parallel trends assumption underlying our Difference-in-Differences design. We do so using the sensitivity analysis of Rambachan and Roth (2023), which evaluates how robust our estimated effects are to potential violations of parallel trends by

<sup>20</sup> This also supports the interpretation that our findings are not driven by unrelated time trends coinciding with subway expansion.

quantifying the magnitude of deviation required to overturn statistical significance. In this framework, the x-axis of Fig. 3 plots  $M$ , the sensitivity parameter that governs the size of allowable deviations. The results for the delinquency indicator remain robust even when permitting pre-trend deviations greater than half a standard error, a benchmark commonly used in recent empirical studies (e.g., Mehic, 2023; Ge et al., 2025). Comparable robustness is observed for other key outcomes, including delinquency amount, prepayment incidence, and prepayment amount. Taken together, these results provide strong reassurance that our findings are not driven by unobserved pre-trends. Full details of the sensitivity analysis are reported in Online Appendix B.

**Confounding Policies.** We also consider whether other macro-level policy changes might confound our results. One potential concern is that major national policy reforms during the study period, such as demonetisation (2016) and the introduction of the Goods and Services Tax (GST, 2017) could influence mortgage performance. However, both policies were implemented nationwide and are absorbed by our year-month fixed effects. Moreover, existing research finds little regional heterogeneity in the effects of demonetisation (Chodorow-Reich et al., 2020; Lahiri, 2020), and GST was applied uniformly within each state (Agarwal et al., 2025). Since our analysis focuses exclusively on Delhi, where intra-state variation is negligible, these reforms are unlikely to bias our estimates.

**Interest Rate Dynamics.** Next, we account for potential confounding from broader financing conditions, particularly the decline in Indian interest rates between 2015 and 2019. During this period, interest rates declined from 8.5% in April 2015 to 5.4% by February 2020. Lower interest rates can stimulate economic activity, leading to lower mortgage delinquency and increased prepayment rates. While we control for time-fixed effects to account for broad interest rate changes that affect all households, there remains a concern that households in postal codes with subway stations might respond more strongly to these rate reductions, potentially skewing our results.

To alleviate this concern, we focus on a sub-sample of households with floating mortgage interest rates. Floating rates are directly tied to market fluctuations, meaning these borrowers should be more responsive to changes in interest rates compared to those with fixed-rate mortgages. If our results are driven primarily by interest rate changes, we would expect to see significant differences in the impact of subway expansions on households with floating rates. However, Table A.8 shows that the results for this sub-sample do not differ significantly from our baseline estimates, indicating that our findings are robust to interest rate fluctuations. Moreover, if interest rates were the primary driver of our results, we would expect the effects to be uniform across postal code areas. Yet, as shown in Table 4, the effects are more pronounced in areas farther from subway stations, suggesting that the primary factor influencing our results is indeed the subway expansions and not interest rate changes.

**Borrower Composition and Unobserved Risk.** Another related concern is that households in treated areas may have lower average loan-to-value (LTV) ratios, which could indicate either greater financial security or larger downpayments, possibly financed through unobserved sources such as black money. To address this, we re-balance the sample by excluding treated households with LTV ratios below the 5th percentile of the treated group. As shown in Table A.9, Panel A, this trimming procedure brings the average LTV across treated and control groups into alignment. Re-estimating our stacked Difference-in-Differences models on this restricted sample (Panel B) yields results consistent with the baseline, suggesting that the observed effects are unlikely to be driven by pre-existing differences in borrower quality or unobserved financing sources.

**Modeling Choices.** Finally, we assess the robustness of our findings to modeling assumptions. Specifically, we examine whether our results

are sensitive to the use of logarithmic transformations for skewed or count-like variables. To address potential distortions from log transformations, we re-estimate our models using the inverse hyperbolic sine (IHS) transformation, which better accommodates zero and small values. As shown in Table A.10, the IHS-based results closely mirror our baseline estimates, reinforcing the consistency of our findings across different functional forms.

Taken together, these robustness checks strengthen the credibility of our identification strategy. Across a range of empirical tests, we find consistent evidence that mortgage performance improves following subway expansions, and that these effects are unlikely to be driven by anticipation, differential pre-trends, national reforms, financing conditions, or modeling choices.

### 5.3. Policy implications

Our study carries direct policy implications, particularly in the context of enhancing financial stability through urban infrastructure investments. While subways are commonly recognized for their role in improving urban mobility and connectivity across cities world-wide (Cheng and Chen, 2015), our findings suggest that their benefits extend beyond transportation. By reducing commuting costs and automobile expenditures, subway expansions can directly influence household financial well-being, particularly in the mortgage market. The ability to reallocate savings from reduced transportation expenses toward mortgage payments not only improves households' financial health but also contributes to greater overall economic stability.

Mortgages represent an important component of household financial portfolios, and difficulties in meeting repayment obligations can have lasting consequences. At the household level, delinquency can lead to ineligibility for public housing assistance and restricted access to credit markets (Collinson et al., 2024), while also negatively affecting overall household welfare (Diamond et al., 2020). At the aggregate level, mortgage delinquencies and foreclosures impose significant costs, potentially exacerbating housing downturns (Campbell et al., 2011; Guren and McQuade, 2020). In contrast, mortgage prepayment benefits both households and the macroeconomy by reducing interest costs and facilitating savings. Prepayments also play a crucial role in the transmission of monetary policy to real economic activity (Beraja et al., 2019).

Against this background, our findings may be particularly relevant for hand-to-mouth households, who tend to have limited liquid assets and are more exposed to short-term financial shocks (Gelman, 2022; Aguiar et al., 2025). For such households, transportation expenses can compete with mortgage obligations for scarce financial resources. In this setting, subway expansions that reduce commuting costs may offer modest but meaningful relief, improving the ability of financially constrained households to meet their mortgage payments.

To further quantify the broader impact of subway expansions on mortgage default rates, we conduct a back-of-the-envelope calculation using prior estimates of foreclosure probabilities. In particular, we estimate the potential reduction in mortgage default rates driven by the observed decrease in delinquency rates following subway expansions. Due to the limitations in our dataset, which does not capture exact default dates for many households beyond our sample period, we rely on this back-of-the-envelope estimation instead of applying the stacked Difference-in-Differences methodology used elsewhere in the paper. Formally, we define  $P(\text{Delinquency}|M)$  and  $P(\text{Delinquency}|-M)$  as the probabilities of delinquency for properties with and without nearby subway stations, respectively. Let  $P(\text{Default}|\text{Delinquency})$  represent the probability of default given delinquency. The reduction in the probability of default due to subway expansions, denoted by  $P(\text{Default}|-M) - P(\text{Default}|M)$ , can then be written as:  $[P(\text{Delinquency}|-M) - P(\text{Delinquency}|M)] \times P(\text{Default}|\text{Delinquency})$ .

$(\text{Delinquency}|M) \times P(\text{Default}|\text{Delinquency})$ .

Our baseline specification estimates the change in the delinquency rate following subway expansions to be  $[P(\text{Delinquency}|M) - P(\text{Delinquency})] = 0.0442$ . To estimate the probability of default given delinquency, we use our loan-level data for February 2020, where  $P(\text{Default}|\text{Delinquency}) \approx 0.019$ .<sup>21</sup> Based on these values, our calculation suggests that the probability of default decreases by:  $0.0442 \times 0.019 \approx 0.008\%$  for households in postal code areas with subway expansions. Out of the 9,681 mortgages in Delhi during our sample period, 72 (0.74%) ended in default. Thus, subway expansions result in an approximate 11% relative decrease in the default rate. These findings demonstrate the potential for infrastructure investments to reduce financial distress. Policymakers should thus consider the far-reaching benefits of urban transport projects, not only in terms of environmental sustainability and mobility but also in supporting financial security for vulnerable households.

While our analysis focuses on short-term mortgage repayment outcomes, the evidence suggests that subway expansions shape both immediate and longer-term financial behavior. In the near term, improved transit access may lead households to delay or forgo vehicle purchases, providing a temporary liquidity boost that supports early mortgage prepayment. Consistent with this mechanism, Fig. 2 shows that the dynamic effects on prepayment are concentrated in the first three months and fade to insignificance after month 7. Over time, however, reduced spending on fuel, maintenance, and parking lowers recurring expenses, enabling households to meet mortgage obligations more consistently and reducing delinquency risk. As Fig. 2 illustrates, the dynamic effects on delinquency are more persistent, remaining significant even after 12 months. More broadly, these patterns indicate that subway expansions strengthen mortgage repayment performance in a sustained way, rather than simply postponing financial distress.

## 6. Conclusion

Our paper highlights an important but often overlooked link between urban infrastructure and household financial health. We show that subway expansions not only promote environmental sustainability by reducing carbon emissions but also improve household balance sheets by lowering transportation costs, thereby enabling more timely mortgage payments. This dual role suggests that public transit investments could be viewed not only as environmental policy but also as an economic tool that strengthens household financial stability and resilience.

Through the phased expansions of the Delhi Metro between 2015 and 2019, we find that new subway stations led to a 4.42% reduction in mortgage delinquency rates and a 1.38% increase in prepayment rates for households in the same postal code. These effects weaken with increasing distance from the station, consistent with localized spillovers. The primary mechanism operates through reductions in automobile expenditures, as financially constrained households substitute away from private vehicle ownership. Vehicle registration data corroborates this channel, showing especially large declines in purchases of lower-quality vehicles, which frees up liquidity for mortgage repayment.

Our findings carry direct policy implications in the context of ongoing global efforts to expand public transportation as part of climate mitigation. The evidence provides a strong rationale for scaling up subway investments, as such projects not only reduce emissions but also alleviate household financial pressures. Back-of-the-envelope calculations indicate that metro expansions could reduce mortgage default rates by 0.08%, yielding meaningful social and economic benefits. More

broadly, these results underscore the importance of aligning transportation investments with urban development and financial inclusion strategies to maximize their environmental and economic returns.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpubeco.2025.105522>.

## Data availability

The data that has been used is confidential.

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<sup>21</sup> Based on our data, there were 2,847 delinquent households as of February 2020. Among these, 53 households ultimately went into default. This method follows Gallagher et al. (2019) who estimate a foreclosure completion probability of 0.18 for delinquent mortgages using Fannie Mae loan performance data.



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